

Q 2 $bx + ay = pq$ — (1)

Sol let $f(x, y, z, p, q) = bx + ay - pq$
Charpit's auxiliary eqn is

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-\frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}}$$

$$= \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} \quad \text{--- (2)}$$

$\Rightarrow \frac{dp}{\dots}$

$$\frac{\partial f}{\partial x} = b, \quad \frac{\partial f}{\partial y} = a, \quad \frac{\partial f}{\partial z} = 0, \quad \frac{\partial f}{\partial p} = x - q$$

$$\frac{\partial f}{\partial q} = y - p$$

Putting all these values in (2)

$$\Rightarrow \frac{dp}{b + p \times 0} = \frac{dq}{a + q \times 0} = \frac{dz}{-b \cdot (x - q) - a \cdot (y - p)}$$

$$= \frac{dx}{-(x - q)} = \frac{dy}{-(y - p)}$$

(taking first two fractions)

$$\Rightarrow \frac{dp}{b} = \frac{dq}{a}$$

$$\Rightarrow \log p = \log q + \log a$$

$$p = aq$$

Putting the value of b in eqn

$$ax + ay = a \cdot r$$

$$x(x+y) = ar$$

$$r = \frac{x(x+y)}{a}$$

$$\Rightarrow b = ar = \cancel{x} \cdot \frac{(x+y)}{\cancel{a}}$$

Putting the value of b & r
in $dz = bdx + rdy$

$$\Rightarrow dz = (x+y)dx + \left(\frac{x(x+y)}{a}\right)dy$$

$$\Rightarrow dz = \frac{(x+y)}{a} (adx + dy)$$

$$\Rightarrow \int dz = \frac{1}{a} \int (x+y)(adx + dy)$$

$$\Rightarrow z = \frac{1}{a} \cdot \frac{(x+y)^2}{2} + b$$

which is the required
complete soln.

4) Solve
 $(b^2 + q^2)y = qz$

Sol $f(x, y, z, b, q) = (b^2 + q^2)y - qz$ — (1)
 Charbit's auxiliary eqn is

$$\frac{dx}{\frac{\partial f}{\partial x} + b \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial q} + q \frac{\partial f}{\partial z}} = \frac{dz}{-b \frac{\partial f}{\partial b} - q \frac{\partial f}{\partial q}}$$

$$= \frac{dx}{-\frac{\partial f}{\partial b}} = \frac{dy}{-\frac{\partial f}{\partial q}} \quad \text{--- (2)}$$

$\Rightarrow \frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 2b^2 + 2q^2, \frac{\partial f}{\partial z} = -q$

$\frac{\partial f}{\partial b} = 2by, \frac{\partial f}{\partial q} = 2qy - z$

Putting all these values in (2)

$$\Rightarrow \frac{db}{0 + by - q} = \frac{dq}{b^2 + q^2 + qy - q}$$

$$= \frac{dz}{-b \cdot (2by) - q(2qy - z)} = \frac{dx}{-2by} = \frac{dy}{-(2qy - z)}$$

$$\Rightarrow \frac{db}{-bq} = \frac{dq}{q^2 - 2b^2y - 2q^2y + qz}$$

$$= \frac{dx}{-2by} = \frac{dy}{-2qy + z}$$

$$\Rightarrow \frac{db}{-bq} = \frac{dq}{bz}$$

$$\Rightarrow bdb + qdr = 0 \Rightarrow b^2 + q^2 = a$$

put $\Rightarrow$ from (i) $ay = qz$

$$\Rightarrow q = \frac{ay}{z}$$

$$b^2 = a - q^2 \Rightarrow b = \pm \sqrt{a - q^2}$$

$$\Rightarrow b = \sqrt{a - \frac{a^2 y^2}{z^2}}$$

$$\Rightarrow b = \sqrt{\frac{az^2 - a^2 y^2}{z^2}} = \frac{\sqrt{az^2 - a^2 y^2}}{z}$$

Putting the value of b & q in
 $dz = bdx + qdy$

$$\Rightarrow dz = \frac{\sqrt{az^2 - a^2 y^2}}{z} dx + \frac{ay}{z} dy$$

$$\Rightarrow z dz = \sqrt{az^2 - a^2 y^2} dx + ay dy$$

$$\Rightarrow \frac{z dz - ay dy}{\sqrt{az^2 - a^2 y^2}} = dx$$

$\Rightarrow$ Multiplying throughout by $2a$.

$$\Rightarrow \frac{2az dz - 2a^2 y dy}{\sqrt{az^2 - a^2 y^2}} = 2a dx$$

$$\Rightarrow \frac{d(az^2 - a^2y^2)}{\sqrt{az^2 - a^2y^2}} = 2adw$$

on integrating,

$$\cancel{2} \sqrt{az^2 - a^2y^2} = \cancel{2} a(x+b)$$

Squaring both sides.

$$az^2 - a^2y^2 = a^2(x+b)^2$$

which is the complete solⁿ.